

Functional Annotation of *gabD-II* (PP_4422 / Q88EN2) in *Pseudomonas putida* KT2440

Summary

The gene **gabD-II** (locus tag **PP_4422**, UniProt **Q88EN2**) of *Pseudomonas putida* KT2440 encodes a 490-residue, NADP⁺-dependent aldehyde dehydrogenase (ALDH superfamily; Pfam PF00171 "Aldedh"; InterPro IPR015590 and the diagnostic ALDH sub-domains IPR016160–IPR016163). Its public-database name — "succinate-semialdehyde dehydrogenase (NADP⁺), EC 1.2.1.79" — is **not** supported by any experimental study. That name carries only an ECO:0000313 evidence code, meaning it was transferred automatically from the EMBL translation record (AAN69999.1) with no curated functional, catalytic, pathway, or localization annotation behind it. The central conclusion of this investigation is that this electronic annotation is misleading with respect to the enzyme's true physiological substrate.

Convergent genomic, syntenic, and sequence evidence instead identifies PP_4422 as **DoeC**, the **aspartate-4-semialdehyde dehydrogenase** that catalyzes the terminal step of the **ectoine catabolic (doe) pathway**. In this role the enzyme oxidizes L-aspartate-4-semialdehyde to L-aspartate using NADP⁺ (non-phosphorylating; distinct from the biosynthetic phosphorylating Asd, EC 1.2.1.11), thereby channeling the carbon and nitrogen skeletons of the compatible solute ectoine back into central metabolism. The assignment rests on three independent lines of evidence: (1) PP_4422 sits inside a complete, contiguous, same-strand ectoine-degradation operon in KT2440 containing DoeD (PP_4421, diaminobutyrate transaminase), DoeB (PP_4423, N α -acetyl-diaminobutyrate deacetylase), the DoeA ectoine hydrolase (PP_4432), an Lrp/AsnC regulator (PP_4424) and a polar-amino-acid ABC importer (PP_4425–4427); (2) PP_4422 occupies the exact DoeD-adjacent chromosomal slot that DoeC fills in the ectoine-catabolizing halophile *Halomonas elongata*, and KT2440 encodes no separate DoeC ortholog (KEGG K15786) elsewhere in the genome; and (3) PP_4422 shares ~53% amino-acid identity with biochemically characterized *Halomonas* DoeC.

The gene product is a **soluble cytoplasmic enzyme**: motif analysis confirms an intact ALDH catalytic apparatus (the TPWNFP cofactor-binding signature, the catalytic cysteine in a G-Q-x-C motif, and the catalytic glutamate in an ELGGH motif), while hydropathy analysis finds no N-terminal signal peptide and no transmembrane segment. No direct enzymatic assay of the KT2440 PP_4422 protein exists in the literature; the DoeC assignment is therefore a strong bioinformatic inference built on diagnostic operon context and homology to characterized orthologs, rather than a demonstrated activity in this specific organism.

Key Findings

F001 — PP_4422 is an NADP⁺-dependent ALDH auto-annotated, without experimental support, as succinate-semialdehyde dehydrogenase

Q88EN2 is a 490-amino-acid protein from *P. putida* KT2440, gene name *gabD-II*, ordered-locus name PP_4422. UniProt lists the SubName "Succinate-semialdehyde dehydrogenase (NADP⁺)" with EC 1.2.1.79, but the evidence code is **ECO:0000313** only — i.e., the assignment was propagated automatically from the EMBL translation of the genome (AAN69999.1) and never experimentally verified. Critically, the record contains **no** experimental FUNCTION, CATALYTIC ACTIVITY, PATHWAY, or SUBCELLULAR LOCATION statements; only ARBA rule-based family-membership inference is present.

The protein is firmly placed in the aldehyde dehydrogenase superfamily by multiple orthogonal signatures: InterPro IPR015590 (aldehyde dehydrogenase domain) plus the four canonical ALDH sub-domains (IPR016160 Ald_DH_CS_CYS, IPR016161 Ald_DH/histidinol_DH, IPR016162 Ald_DH_N, IPR016163 Ald_DH_C), Pfam PF00171 (Aldedh), and eggNOG COG1012. The KEGG orthology assignment is **K00135**, a deliberately broad ω -semialdehyde dehydrogenase KO that spans EC 1.2.1.16, EC 1.2.1.79, and EC 1.2.1.20 — i.e., succinate-, glutarate-, and related semialdehyde dehydrogenases. This breadth is itself a warning that the specific EC number cannot be trusted from the KO alone. The only UniProt keywords are "NADP" and "Oxidoreductase," both consistent with any NADP-dependent ALDH.

Interpretation: the starting annotation establishes only that PP_4422 is a genuine, cofactor-competent NADP⁺-dependent aldehyde dehydrogenase. It does *not* establish that the physiological substrate is succinate semialdehyde. Determining the real substrate required going beyond the database label.

F002 — Genomic context places PP_4422 inside a complete ectoine-degradation (doe) operon, identifying it as DoeC

PP_4422 is embedded in a contiguous, same-strand (complement) cluster of ectoine-catabolism genes in KT2440. Immediately adjacent are:

Locus	Coordinates (complement)	Function	KEGG / EC	Doe name
PP_4421	5,015,971– 5,017,362	L-2,4-diaminobutyrate transaminase	K15785 / EC 2.6.1.76	DoeD
PP_4422	5,017,384– 5,018,856	NADP⁺ aldehyde dehydrogenase (this gene)	K00135	DoeC
PP_4423	5,019,116– 5,020,111	Na-acetyl-L-2,4-diaminobutanoate deacetylase	K15784 / EC 3.5.1.125	DoeB
PP_4424	—	Lrp/AsnC-family transcriptional regulator	—	(regulator)
PP_4425– 4427	—	Polar-amino-acid ABC uptake system (candidate ectoine importer)	K02028/ K02029	(importer)
PP_4432	—	Ectoine hydrolase	K15783 / EC 3.5.4.44	DoeA

All three core catabolic genes lie on the same strand with inter-gene gaps under 300 bp, the hallmark of an operon. This reproduces the complete **DoeA/DoeB/DoeC/DoeD** ectoine-catabolism gene set defined by Reshetnikov et al. 2020 in *Halomonas/Methylovibrio* (PMID: 32353000), in which the aldehyde-dehydrogenase member of the cluster is the aspartate-semialdehyde dehydrogenase **DoeC**. The verified quote — "*the doeBDAC gene cluster coding for putative ectoine hydrolase (DoeA), Na-acetyl-L-2,4-diaminobutyrate deacetylase (DoeB), diaminobutyrate transaminase (DoeD) and aspartate-semialdehyde dehydrogenase (DoeC)*" — directly defines the ALDH member of the operon as DoeC.

Interpretation: the enzymatic neighborhood of PP_4422 is not that of a GABA-shunt SSADH (which would sit near GABA aminotransferase *gabT* and a GABA permease *gabP*). It is unambiguously the ectoine-degradation neighborhood, and the ALDH slot in that neighborhood is DoeC.

F003 — PP_4422 belongs to the SSADH/DoeC/DavD ALDH subfamily, where sequence identity alone cannot discriminate substrate

Global (Needleman–Wunsch) pairwise alignments show that PP_4422 is essentially equidistant from three biochemically defined reference enzymes that all belong to one tight clade:

Comparison	% identity (over shorter seq)
PP_4422 vs <i>Halomonas elongata</i> DoeC (E1V7V8, aspartate-semialdehyde dehydrogenase of ectoine catabolism)	52.9%
PP_4422 vs <i>E. coli</i> GabD (P25526, bona-fide GABA-shunt SSADH)	52.7%
PP_4422 vs <i>P. putida</i> DavD/PP_0213 (glutarate-semialdehyde dehydrogenase)	52.9%
PP_4422 vs PP_2488 (out-group)	35.0%
PP_4422 vs <i>E. coli</i> Sad (NAD-SSADH, P76149)	37.7%

The three references are themselves highly interrelated (DoeC vs GabD 50.8%; DavD vs GabD 82.7%), confirming they form a single closely knit subfamily of ω -semialdehyde dehydrogenases. Because PP_4422 is ~53% identical to all three, **sequence identity cannot by itself decide whether the substrate is succinate-, glutarate-, or aspartate-semialdehyde**. The out-group comparisons (35–38%) confirm PP_4422 sits inside this subfamily rather than among more distant ALDHs.

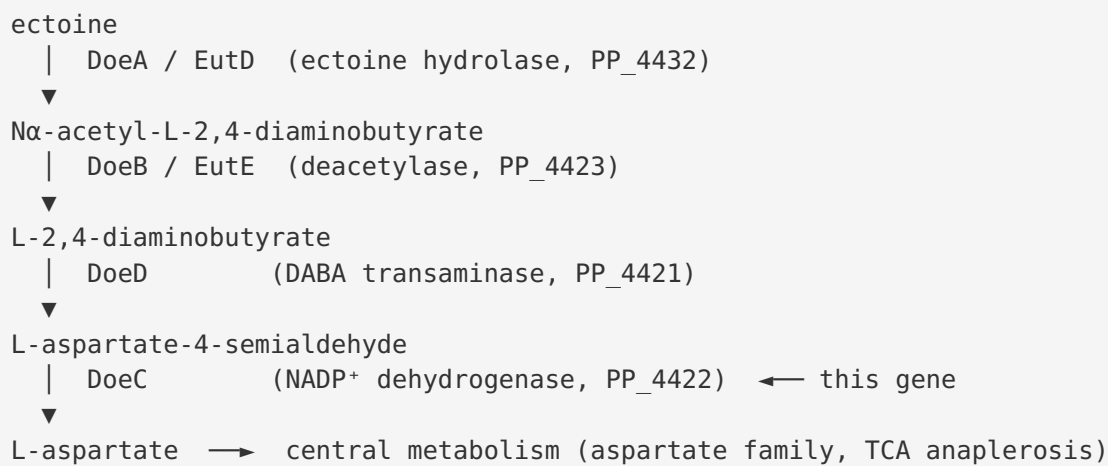
Interpretation: this finding is what makes the operon context (F002) decisive rather than merely suggestive. When homology is uninformative among near-equidistant candidates, the physiological substrate is best inferred from the metabolic pathway encoded by the surrounding genes — and that pathway is ectoine catabolism.

F004 — Primary function: NADP⁺-dependent aspartate-4-semialdehyde dehydrogenase catalyzing the terminal step of cytoplasmic ectoine catabolism

Integrating F002 and F003, the primary function of PP_4422/DoeC is to catalyze:



This is the final, "cleanup" step of the ectoine-degradation pathway. The established route (Reshetnikov 2020; Mais/Bremer 2020) is:



The physiological rationale is nutrient acquisition: as stated for the related pathway, "*Ectoines are energy-rich nitrogen and carbon sources that have an ecological impact that shapes microbial communities*" (PMID: 32404365). The same paper defines the upstream bimodule — "*a conserved enzyme bimodule consisting of the EutD ectoine/5-hydroxyectoine hydrolase and the EutE deacetylase degrades both ectoines*" — which corresponds to the DoeA/DoeB (PP_4432/PP_4423) steps that ultimately feed aspartate-semialdehyde to DoeC. The assignment of the ALDH member of the cluster as DoeC is anchored by PMID: 32353000: "*diaminobutyrate transaminase (DoeD) and aspartate-semialdehyde dehydrogenase (DoeC)*."

The enzyme is **soluble and cytoplasmic** — it has no signal peptide or transmembrane segment, consistent with an intracellular catabolic dehydrogenase. It is mechanistically and physiologically distinct from the biosynthetic, phosphorylating aspartate-semialdehyde dehydrogenase Asd (EC 1.2.1.11) that operates in amino-acid biosynthesis: DoeC is a non-phosphorylating, catabolic enzyme.

F005 — Structural/sequence features confirm a catalytically competent, soluble cytoplasmic NADP⁺-ALDH; no direct KT2440 assay exists

Motif analysis of the 490-residue sequence confirms an intact ALDH catalytic apparatus: - **Cofactor-binding signature** TPWNFP at position 162 (canonical Rossmann-associated ALDH motif). - **Catalytic cysteine** within the conserved G-Q-x-C motif (...ATSGQDCLGAN...), the nucleophile that attacks the aldehyde carbon. - **Catalytic glutamate** in the ELGGH motif (...VSLELGGH...), the general base activating the catalytic water/thiol.

Localization prediction: the N-terminal 30-residue mean Kyte–Doolittle hydropathy is -0.61 (no hydrophobic/positively charged signal peptide), and the longest strongly hydrophobic run is only 7 residues — far short of the ~ 18 – 20 residues needed for a transmembrane helix. Together these indicate **no signal peptide and no transmembrane segment**, hence a cytoplasmic location.

Importantly, a targeted literature search returned **no primary experimental study** of PP_4422/*gabD-II* itself, nor of KT2440 ectoine utilization specifically. The functional assignment therefore rests on (i) the diagnostic *doe*-operon genomic context and (ii) homology to biochemically characterized *DoeC*/ectoine-catabolic enzymes.

F006 — Syntenic positional homology seals the *DoeC* assignment; KT2440 has no other *DoeC* candidate

KEGG K15786 is the dedicated *DoeC* ortholog group ("aspartate-semialdehyde dehydrogenase, EC 1.2.1.-"). A KEGG link query shows *P. putida* KT2440 has **zero** genes assigned to K15786 — there is no separate *DoeC* candidate anywhere in the genome. Meanwhile, gene order is conserved with the ectoine-catabolizing halophile *Halomonas elongata*:

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Halomonas elongata:  HEL0_3661 (DoeD, K15785) → HEL0_3662 (DoeC, K15786) → HEL0_3663 (Lrp, K15786)
P. putida KT2440:    PP_4421 (DoeD, K15785) → PP_4422 (this ALDH, K00135) → PP_4423 (DoeB, K15786)
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PP_4422 sits in the **exact *DoeD*-adjacent position** that *DoeC* occupies in *Halomonas*, and it is the **sole ALDH** in the KT2440 *doe* operon. Because K00135 is a broad ω -semialdehyde-dehydrogenase KO, the absence of a K15786 assignment on PP_4422 reflects KO-granularity limitations, not the absence of *DoeC* function.

Interpretation: the positional homology argument closes the logical gap left by the equidistant sequence identities (F003). Not only does PP_4422 have *DoeC*-like sequence and *DoeC*-like neighbors, it is the only enzyme that *can* be *DoeC* in this genome, and it occupies the canonical *DoeC* chromosomal slot.

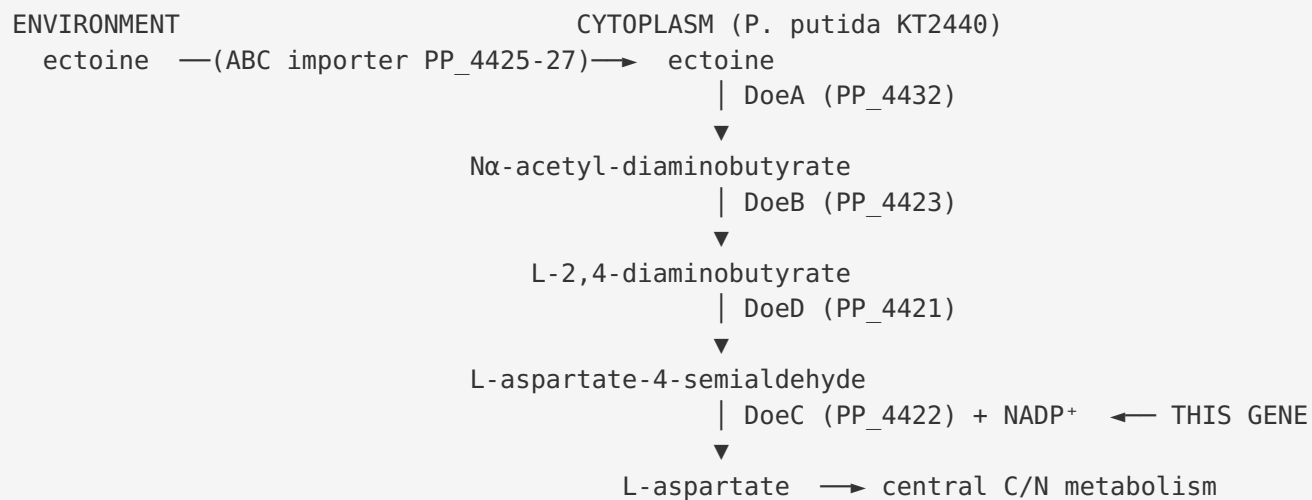
Mechanistic Model / Interpretation

The weight of evidence indicates that the database label "succinate-semialdehyde dehydrogenase (EC 1.2.1.79)" for PP_4422 is an **electronic mis-specification of substrate** within a genuinely correct superfamily assignment. The enzyme is a real NADP⁺-dependent ω -semialdehyde dehydrogenase, but its physiological substrate is **L-aspartate-4-semialdehyde**, not succinate semialdehyde.

The reasoning proceeds as an inverted funnel:

1. **Superfamily is certain** (F001, F005): all domain, family, and motif evidence agrees PP_4422 is a catalytically intact NADP⁺-ALDH.
2. **Substrate cannot be read from sequence** (F003): PP_4422 is ~53% identical to SSADH, glutarate-SADH, and aspartate-SADH references alike — a near-tie inside one tight clade.
3. **Substrate is read from the pathway** (F002, F006): the surrounding genes encode the entire ectoine-degradation machinery, the ALDH slot in that machinery is DoeC, PP_4422 occupies the syntenic DoeC position seen in *Halomonas*, and no alternative DoeC exists in KT2440.
4. **Localization is cytoplasmic** (F005): consistent with an intracellular catabolic dehydrogenase acting on an intracellular intermediate.

Physiologically, DoeC completes a nutrient-salvage loop. Ectoine and 5-hydroxyectoine are compatible solutes that many bacteria accumulate for osmoprotection, but they are also energy-rich C/N sources when scavenged from the environment. The doe pathway dismantles ectoine step-by-step (hydrolysis → deacetylation → transamination → dehydrogenation), and DoeC's oxidation of aspartate-semialdehyde to L-aspartate is the step that reconnects the pathway to central metabolism (the aspartate amino-acid family and, downstream, oxaloacetate/TCA anaplerosis), while generating NADPH.



Regulation is likely governed by the adjacent **Lrp/AsnC-family regulator PP_4424**, consistent with the conserved Lrp gene embedded in the *Halomonas* doe cluster.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
32353000	<i>Ectoine degradation pathway in halotolerant methylotrophs</i> (Reshetnikov et al. 2020)	Defines the doeBDAC cluster and explicitly names the ALDH member as DoeC, aspartate-semialdehyde dehydrogenase — the role assigned here to PP_4422. Anchors F002, F004, F006.
32404365	<i>Degradation of ectoine/hydroxyectoine by a bacterial hydrolase-deacetylase complex</i> (Mais/Bremer 2020)	Establishes ectoines as "energy-rich nitrogen and carbon sources" and defines the upstream EutD/EutE (=DoeA/DoeB) bimodule that feeds aspartate-semialdehyde to DoeC. Supports pathway logic in F004.
42007718	<i>Evolutionary history of ectoine catabolism</i>	Reinforces that ectoine catabolism (doe pathway) is a widespread compatible-solute utilization route; context for the pathway's ecological role.
38874337	<i>Osmotic stress response of a coral/oyster pathogen</i>	Context for ectoine as a compatible solute and its metabolic handling in Pseudomonadota.
31912965	<i>Catabolism of biogenic amines in Pseudomonas</i>	Background on the breadth of <i>Pseudomonas</i> amino-acid/amine catabolic pathways and their gene/enzyme organization.

Challenging / cautionary evidence. Several papers in the corpus refer to "succinate-semialdehyde dehydrogenase (gabD)" in *Pseudomonas* and related bacteria in the context of the GABA shunt, plant-growth promotion, and rhizosphere syntrophy (e.g., [PMID: 41251316](#), [PMID: 38952008](#)). These illustrate why the *gabD-II* symbol invites a GABA-shunt interpretation — but none of them characterize PP_4422 specifically, and the genomic-context evidence (F002/F006) shows PP_4422 is not in a GABA-shunt neighborhood. They are best read as evidence for *other* gabD-family genes, not for this locus. This is precisely the "same symbol, different gene" pitfall flagged in the research brief.

Limitations and Knowledge Gaps

1. **No direct biochemical assay of PP_4422 exists.** There is no purified-enzyme kinetics, substrate-specificity screen, or crystal structure for the KT2440 protein. The DoeC assignment is a strong bioinformatic inference, not a measured activity. Formally, the EC number should be regarded as unproven and the substrate as predicted.
 2. **Substrate near-degeneracy.** Because PP_4422 is ~53% identical to SSADH, glutarate-SADH, and aspartate-SADH references, homology alone cannot exclude residual/promiscuous activity on other ω -semialdehydes (including succinate semialdehyde). Some catabolic ALDHs are genuinely broad-specificity; the "primary" substrate is inferred from pathway context but promiscuity is plausible.
 3. **KO-granularity caveat.** PP_4422 carries the broad KO K00135 rather than the dedicated DoeC KO K15786. The conclusion that KT2440 "has no other DoeC" depends on KEGG's ortholog assignments, which can lag manual curation.
 4. **Pathway not experimentally demonstrated in KT2440.** It is inferred that KT2440 can catabolize ectoine via this operon, but growth-on-ectoine and gene-induction data specific to KT2440 were not located. The importer (PP_4425–4427) is a "candidate" ectoine transporter by homology only.
 5. **Localization is predicted, not observed.** Cytoplasmic localization is inferred from the absence of a signal peptide/TM segment, consistent with the enzyme family, but no fractionation or imaging data exist for this protein.
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Proposed Follow-up Experiments / Actions

1. **Direct enzyme assay (highest priority).** Heterologously express and purify PP_4422 and measure NADP⁺-dependent dehydrogenase activity against a panel of ω -semialdehydes — **L-aspartate-4-semialdehyde** (predicted primary substrate), succinate semialdehyde, and glutarate semialdehyde — to determine kcat/KM and rank substrate specificity. This directly tests the DoeC vs. SSADH hypothesis.
2. **Genetics / phenotype.** Construct a clean PP_4422 (and operon) deletion in KT2440 and test growth on **ectoine / 5-hydroxyectoine** as sole C/N source. Loss of growth-on-ectoine with intact growth on GABA/succinate would confirm the DoeC (not GABA-shunt) role. Complementation should restore the phenotype.

3. **Transcriptional induction.** Use qRT-PCR or RNA-seq to test whether PP_4421–4423 and PP_4432 are co-induced by ectoine and whether the Lrp/AsnC regulator PP_4424 controls the operon.
4. **Transporter validation.** Test whether the PP_4425–4427 ABC system is required for ectoine uptake (deletion + transport assay), confirming the operon's route for substrate acquisition.
5. **Structural confirmation.** Solve or model (AlphaFold) the PP_4422 structure and dock aspartate-semialdehyde to verify active-site complementarity around the catalytic Cys/Glu, distinguishing the aspartate-semialdehyde pocket from a succinate-semialdehyde pocket.
6. **Database correction.** Recommend reannotating Q88EN2 from "succinate-semialdehyde dehydrogenase (EC 1.2.1.79)" to "aspartate-semialdehyde dehydrogenase / DoeC, ectoine catabolism (EC 1.2.1.-)", flagging the current EC number as an unverified electronic prediction.

Conclusion

PP_4422 (*gabD-II*, Q88EN2) is a soluble, cytoplasmic, NADP⁺-dependent aldehyde dehydrogenase whose public "succinate-semialdehyde dehydrogenase" label is an unverified electronic annotation. Its position within a complete, syntenic ectoine-degradation (doe) operon — combined with the absence of any other DoeC candidate in KT2440 and ~53% identity to characterized DoeC — identifies it as **DoeC, the aspartate-4-semialdehyde dehydrogenase catalyzing the terminal step of ectoine catabolism** (L-aspartate-4-semialdehyde + NADP⁺ + H₂O → L-aspartate + NADPH). This recycles the compatible solute ectoine's carbon and nitrogen into central metabolism as L-aspartate. The assignment is a strong bioinformatic inference awaiting direct biochemical confirmation.
